Exp No:13.a Conduct an experiment to show data visualization using line plot

CODE:

```
import numpy as np

import pandas as pd

df = pd.DataFrame({

   'Country':
['India','Morocco','Germany','Morocco','Germany','India','Morocco','I
ndia','Germany','India'],

   'Age': [44, 27, 30, 38, 40, 35, np.nan, 48, 50, 37],

   'Salary': [72000, 48000, 54000, 61000, np.nan, 58000, 52000,
79000, 83000, 67000],

   'Purchased': ['No', 'Yes', 'No', 'No', 'Yes', 'Yes',

            'No', 'Yes', 'No', 'Yes']

})

print("Original Dataset:")

print(df)

print("\n\nDataFrame Information:")

print(df.info())

print("\n\nCountry Mode:")

print(df.Country.mode())

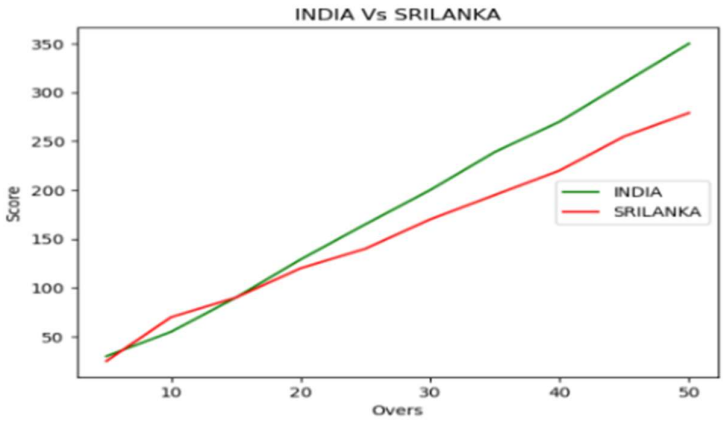
print("\n\nFirst Mode Value:")

print(df.Country.mode()[0])

df['Country'] = df['Country'].fillna(df.Country.mode()[0])
```

```
df['Age'] = df['Age'].fillna(df.Age.median())
df['Salary'] = df['Salary'].fillna(round(df.Salary.mean()))
print("\n\nDataset After Filling Missing Values:")
print(df)
print("\n\nOne Hot Encoding:")
print(pd.get_dummies(df.Country))
updated_dataset = pd.concat(
   [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
   axis=1
)
print("\n\nUpdated Dataset:")
print(updated_dataset)
print("\n\nDataFrame Information After Preprocessing:")
print(df.info())
updated_dataset['Purchased'] =
updated_dataset['Purchased'].replace(
   ['No', 'Yes'], [0, 1]
)
print("\n\nFinal Dataset:")
print(updated_dataset)
```

OUTPUT:



RESULT: data visualization using line plot has been successfully verified.

Exp No:13.b Conduct an experiment to show data visualization using bar chart

CODE

```
import matplotlib.pyplot as hscmark

import numpy as np

Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']

xaxis = np.arange(len(Names))

Percentage_hsc = [96, 91, 94, 75, 45, 81]

hscmark.bar(Names, Percentage_hsc)

hscmark.xticks(xaxis, Names, rotation=45)

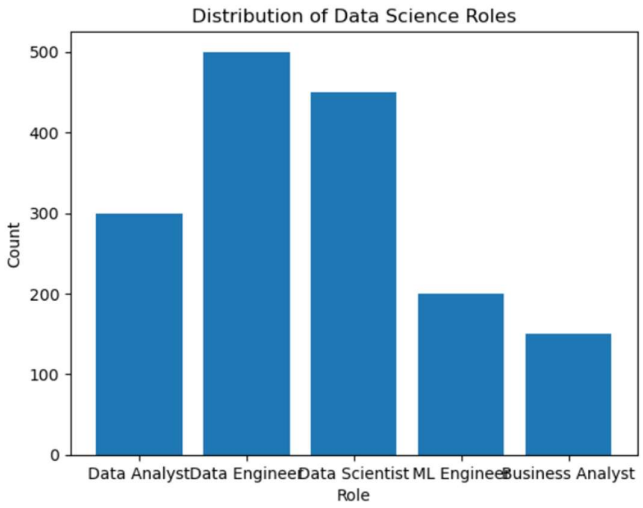
hscmark.xlabel("Names of Pupil")

hscmark.ylabel("Percentage")

hscmark.title("Comparison of HSC Percentage", fontsize=20,
color="green")

hscmark.show()
```

OUTPUT



RESULT: hence data visualization using bar chart has been represented successfully.

Exp No:13.c Conduct an experiment to show data visualization using pie chart

```
import matplotlib.pyplot as election

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']

Votes = [315, 130, 245, 210]

colors = ['green', 'yellow', 'red', 'orange']

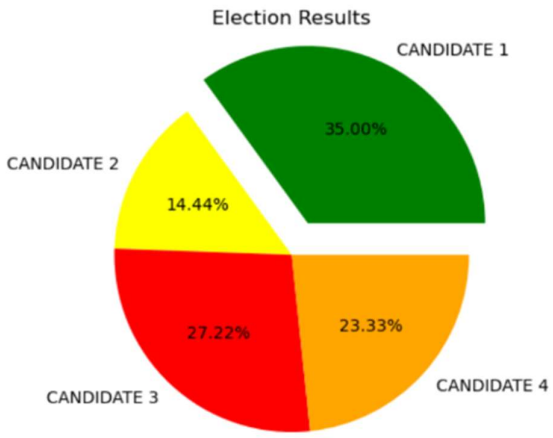
explode = (0.2, 0, 0, 0)

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')

election.title('Election Results')

election.show()
```

OUTPUT



RESULT: data visualization using pie chart has been successfully verified.